

Executive Summary

The main aim of this study is modelling the concentration of pollutant at a given distance away from a waste incinerator at Renfrew recycling centre, the South Clyde Energy Project, assuming a constant pollution source. They would like to know the optimal method and time for releasing pollutant such that it has minimal effect on the community. Since the site is located west of Glasgow city centre we consider prevailing winds in direction Northeast carrying pollutants over the city. We aim to lower pollution early and assessed pollutant concentration 8 km downwind of the source (residential area). Our model is based on analytically and numerically solving a 1D advection-diffusion equation with constant and varying advection term using standard diffusion coefficient for various pollutant-release strategies and data-fitted sine-wave advection modelling wind velocity. The resulting model incorporates a solver class which allows flexible parameter-based testing of different strategies. We then considered four release strategies: instantaneous, regular bursts, uniform continuous, and inverse-wind release. This study determines that continuous release of pollutants is optimal. For future studies of this topic, a higher dimensional model, incorporating advection, is recommended as this can better account for downwind pollution near ground level and localised pollution. It is further recommended to conduct research on how chemical reactions affect the dispersion of pollutants in dry and wet conditions, such that models can be refined for different pollutants and wind patterns in future studies.

1 The Model

We choose to model the transport of air pollutants using a one-dimensional diffusion–advection equation defined over an infinite spatial domain. Since our primary focus is the effect of wind strength and direction, an advection-led equation is preferred, similar models have also been applied before [8]. The 1D advection-diffusion equation is given by,

$$P_t(x, t) = DP_{xx}(x, t) + vP_x(x, t) \quad \text{with} \quad x \in (-\infty, \infty), \quad t \geq 0, \quad (1)$$

for pollution concentration $P(x, t)$, diffusion coefficient D , which depends on chemical properties of the pollutant, and advection term v representing varying wind speeds. We fix a constant diffusion coefficient and considered both constant and time-dependent choices of v . Our model has two variables, the horizontal distance $-\infty < x < \infty$ and time $t \geq 0$, where our spatial coordinate represents a horizontal distance at a fixed height, corresponding to the height of the chimney.

We start by considering our model to be subject to the initial condition $P(x, 0) = \delta(x)$, the Dirac delta function. This ‘infinite spike’ at $x = 0$ models an instantaneous pollutant release from our chimney. We use Dirichlet boundary conditions $P(x, t) \rightarrow 0$ as $x \rightarrow \pm\infty$ representing negligible pollutant concentration far away from the incinerator.

2 Analytical solution

Analytically, we solve the equation [1] for constant v and initial and boundary conditions,

$$P(x, 0) = \delta(x), \quad P(x, t) \rightarrow 0 \quad \text{as} \quad x \rightarrow \pm\infty.$$

We perform a change of variables, $\eta = x - vt$, reducing the advection–diffusion equation to a 1D diffusion equation $P(x, t) = U(x - vt, t)$ ¹, where $U(x, t)$ satisfies

$$U_t(x, t) = DU_{xx}(x, t) \quad \text{s.t.} \quad U(x, t) \rightarrow 0 \quad \text{as} \quad x \rightarrow \pm\infty. \quad (2)$$

We solve equation [2] using the Fourier transform. The definition of the forward and inverse Fourier transforms for a continuous function $f(x)$ are [1],

$$\mathcal{F}[f(x)] = \hat{f}(k) = \int_{-\infty}^{\infty} f(x)e^{-ikx} dx \quad \& \quad \mathcal{F}^{-1}[\hat{f}(k)] = f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(k)e^{ikx} dk \quad (3)$$

Using Fourier transform of derivatives through integration by parts, $\mathcal{F}\left[\frac{\partial^n U}{\partial x^n}\right] = (ik)^n \mathcal{F}[U(x, t)]$ [1], our equation becomes,

$$\mathcal{F}[U_t(x, t)] = \mathcal{F}[DU_{xx}(x, t)] \implies \hat{U}_t(k, t) = -Dk^2 \hat{U}(k, t), \quad (4)$$

where $\mathcal{F}[U(x, t)] = \hat{U}(k, t)$. This is a first-order ODE we can solve using the integrating factor $e^{-Dk^2 t}$ to achieve the solution,

$$\hat{U}(k, t) = \hat{U}(k, 0)e^{-Dk^2 t}. \quad (5)$$

¹A result we discussed in the lectures involving constant advection.

Recall, our initial condition is given by $U(x, 0) = P(x, 0) = \delta(x)$. Since the Dirac delta function satisfies $\int_{-\infty}^{\infty} \delta(x) dx = 1$ and by use of the sifting property [1], we apply the Fourier transform to the initial condition to achieve $\hat{U}(k, 0) = 1$. We then take the inverse Fourier transform of Equation [4] and integrate to achieve,

$$U(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-Dk^2 t} e^{-ikx} dk = \frac{e^{-\frac{x^2}{4Dt}}}{\sqrt{4\pi Dt}}. \quad (6)$$

We can then find the solution to Equation [1] by reversing the change of variables. Thus, our analytical solution is given by,

$$P(x, t) = \frac{1}{\sqrt{4\pi Dt}} e^{-\frac{(x-vt)^2}{4Dt}}. \quad (7)$$

3 Numerical Solution

For realistic computation we truncate the spatial interval to $x \in [-L, L]$, the time interval to $t \in [0, T]$ with $T = 24$ hours and $L = 600km$ (effectively infinite). The total daily burst is set as $k = 1kg/km$, consistent with typical reported values [6]. To solve PDE (1) numerically we first non-dimensionalise and then use finite differencing schemes varying $v(t)$ at each time step. We discretize the diffusion term in space using centred difference. To discretize the advection term, we used an upwind scheme which switches method of differencing depending on the direction of advection flow. ie. if $v > 0$ we used forward finite difference and if $v < 0$ we use a backward finite difference scheme. Finally, we discretize time using implicit Euler due to structural stability [7]. Implicit Euler allows for far more parametric flexibility than explicit Euler as it has higher range of stable parameters given physical damping threshold (usually 1). We note it can still be numerically unstable in some cases, hence we ensure our parameters are selected in coherence with stability, such as avoid taking a very low D value. For our initial comparison against the analytical solution, we used constant unit advection and diffusivity ($v = 1, D = 1$) in the case of an instantaneous burst of pollution (see Figure [1]). The shape of both plots grow increasingly close, suggesting numerical accuracy in reflecting overall trend. Secondly, numerical peak remains above analytical peak, suggesting little numerical diffusion error.

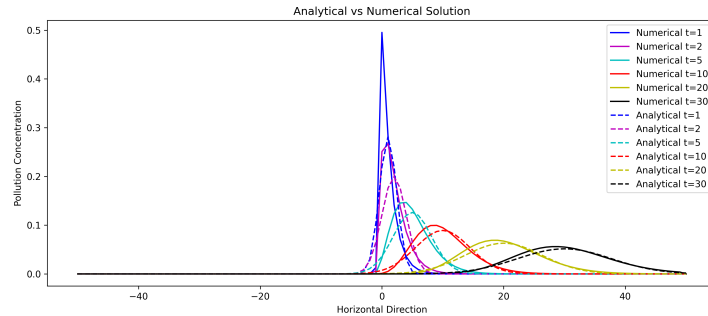


Figure 1: Comparing our analytical and numerical solution.

There are two concerns but they are negligible. Firstly, the deviation for $t=1$ is expected due to the numerical approximation of the initial Dirac delta condition. Secondly, numerical peak to the left of analytical peak suggests some dispersion error, however, since we are more interested in the shape than concentration accuracy due to data limitations anyway, it is negligible.

4 Wind Variation

To model realistic wind variation, hourly wind-speed data located at the South Clyde Energy Centre (2005–2024) was obtained via Open Meteo [2]. This was processed to remove potentially distorting entries, such as leap-day measurements. Each observation was decomposed into u and v components representing wind velocity in the East and North directions respectively. Data from 2015–2024 was used for model training to balance data quantity with recency, due to changing wind speeds over time [3], while data from 2005–2014 serves as an independent test set to evaluate model generalizability. Exploratory data analysis revealed significant monthly variations in average hourly wind patterns in both u , v components. We developed a sinusoidal model incorporating three harmonics for annual, daily, and half-daily wind cycles. The model and its performance evaluations are provided in the submitted code. Although the unified model better evaluates seasonal transitions, our analysis focused on the summer model for its larger intra-day wind variation which affects emissions the most. Hence, we built a separate summer/winter model using a double-harmonic fit to measure daily and half-daily wind cycles which captures a significant proportion of daily variance [4]. The model function f with fitted parameters in the summer season is

$$f(t) = A_1 \sin(2\pi t/24) + B_1 \cos(2\pi t/24) + A_2 \sin(2\pi t/12) + B_2 \cos(2\pi t/12) + C.$$

	A_1	B_1	A_2	B_2	C
u	-1.047	-1.308	0.459	0.073	3.365
v	-0.098	-0.545	0.066	0.336	2.464

Table 1: Fitted parameter values for each component in the summer season

Prevailing winds at the site are predominantly Northeast (from the coast), so an average of u and v components were used to represent the prevailing wind speed, giving a final wind-speed function

$$V(t) = \frac{u(t) + v(t)}{\sqrt{2}}.$$

Comparing simulated data with test data revealed strong agreement in curve shapes. We observe that in spite of the slight underprediction in the summer model we have a high Pearson correlation coefficient of 0.988 and a low normalised Root Mean Square Error (nRMSE) of 0.113 indicating realistic wind speed estimates. Furthermore, the model fits well in both amplitude and phase which supports our focus on the wind profile’s shape.

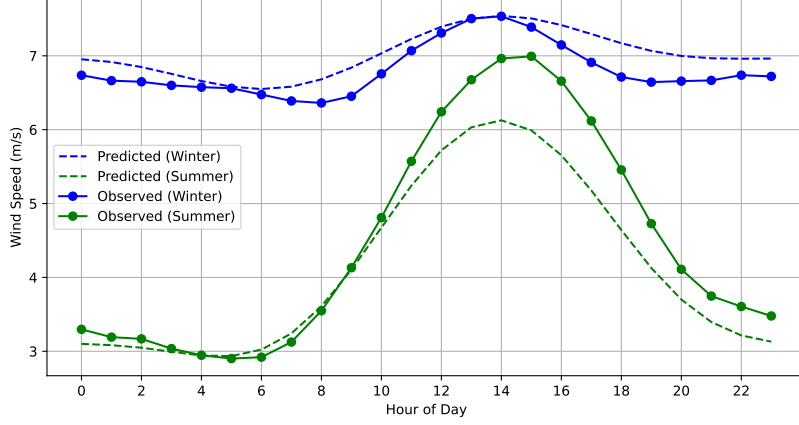


Figure 2: Fitted vs Test: Average hourly NE wind speed by season

5 Results and Discussion

We modelled four pollution release strategies over a 24-hour period: an instantaneous burst, a continuous uniform release, twelve regular bursts, and an inverse-wind method (polluting most at low wind and least at high wind). After testing for instantaneous burst, we conclude wind carries pollutants away after 2 hours, hence we decide to conduct bursts at 2-hour intervals. We applied our wind model and a diffusion coefficient $D = 0.36 km^2/h$, based on commonly observed values [5]. Results for each method are shown in Figure [3].

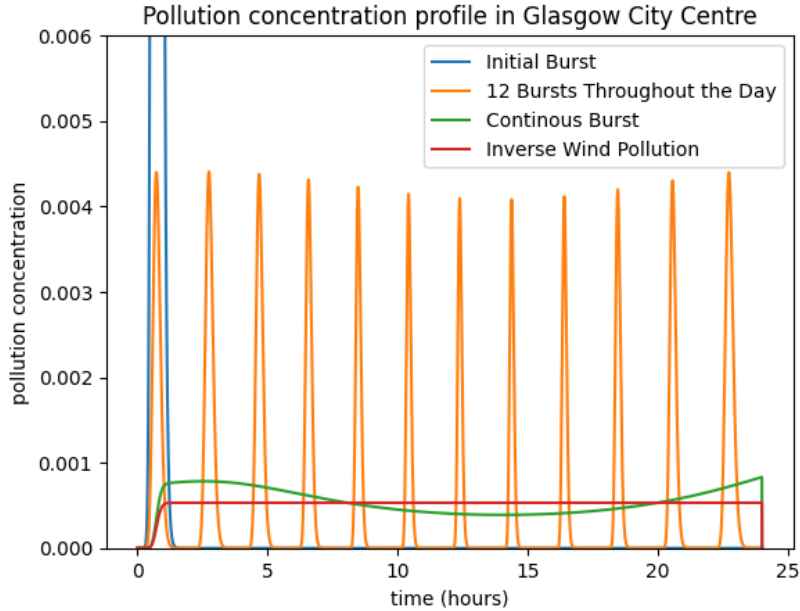


Figure 3: Pollution concentration profile 8km downwind of the source - located at the centre of Glasgow.

Although quickly lowered, instantaneous and burst method results in singular concentration

peaks constituting possible health concerns and likely exceeding city pollution threshold. The continuous and inverse-wind releases perform best with consistently low concentration levels. Although both appear stable, a build up of pollution levels under low wind conditions could pose concerns near the source. The continuous-release model, though higher in the early hours, remains lower during the day which is optimal for citizens and is therefore the recommended strategy.

6 Further work and 2D Model

To better explain pollution behaviour over a region, we extend the framework in (1) to two spatial dimensions. Ideally, it includes advection to model directional transport with an initial velocity flux field, representing the full 2D advection–diffusion process. However, this is beyond the project. Here, we focus on a simplified 2D diffusion model for concentration $P(x, y, t)$,

$$P_t(x, y, t) = D (P_{xx}(x, y, t) + P_{yy}(x, y, t)) . \quad (8)$$

Boundary and initial conditions follow (1), adapted to 2D. We approximate the PDE again using implicit Euler finite-differencing scheme and generate a system of matrix equations, which is then solved numerically. Results are shown in Figure [4].

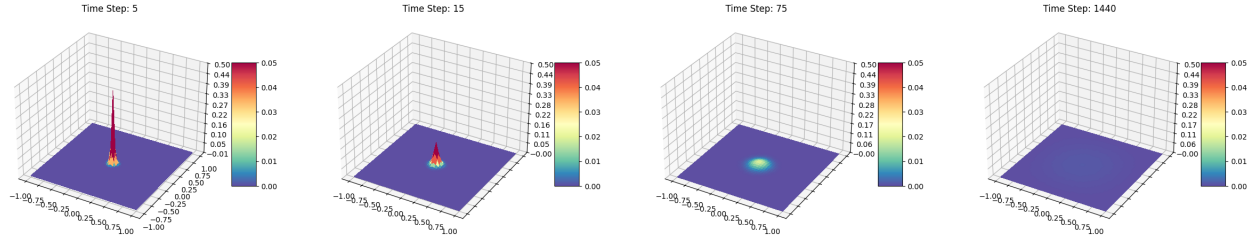


Figure 4: Left to right: Numerical solution of the 2D diffusion equation for increasing time steps.

7 Conclusion

We conclude the optimal pollutant release strategy is a continuous burst starting at midnight. The model indicates slightly higher pollution levels at night due to fluctuating wind speeds but keeps daytime levels low, reducing exposure for Glasgow residents. Future work should focus on extending the model to higher dimensions to account for varying wind directions and near-ground pollution, as well as investigating how dry and wet conditions influence pollutant dispersion.

References

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Reflective Report

Throughout this project, our group developed significantly in terms of teamwork, communication, and project management. Although the process presented several challenges, these provided valuable opportunities for learning that have enhanced both our academic and professional capabilities.

At the start, we focused on dividing the tasks according to individual strengths, with the aim of balancing the workload. However, we quickly realised that the development of an analytical solution could not be evenly distributed without unnecessary duplication of effort. This taught us that efficient delegation is not simply about fairness but rather how to balance skill-sets with task complexity. In future work, we will not over subscribe people to jobs and could've assigned people to read widely or begin further exploratory work in order to gain deeper understanding of the problem.

Another key learning point was the importance of a shared understanding of the overall direction of the project as we found that our individual research tasks were often interlinked. For example, the analysis of wind data and development of a suitable model to predict wind-speeds needed to be completed before we could implement and conduct tests with our advection-diffusion model. Although we worked diligently on individual components, at times we lacked a unified view of how each element contributed to the final model. To counteract this, we eventually implemented weekly progress updates, where each person briefly presented progress, challenges, and next steps, which helped bridge this gap. In hindsight, beginning this structured reflection earlier would have improved group cohesion and alignment.

The process of combining individual work proved to be one of our biggest challenges throughout the project, with work often overlapping. To address this, we implemented a shared GitHub repository to facilitate version control, which significantly improved coordination and transparency across the group. However, managing a single shared Jupyter notebook with six contributors led to frequent merge conflicts. In future work, we plan to create personal branches for each team member, to allow for exploratory work before merging key contributions. This more closely reflects industry standard collaborative practices.

From an employability perspective, this project helped us develop many skills that are highly valued by employers, including time-management, problem-solving, teamwork, and communication. Our experience resolving conflicts in GitHub emphasised the importance of structured collaboration and the need to use version control systems effectively. While the development of our numerical models in Python clearly enhanced our coding and modelling skills, the greatest value came from the experience of conducting research and effectively integrating that research into the work of the team. The delegation of sub-tasks and the interdependence of these tasks helped to develop our sense of accountability, as each member was responsible not only for completing their own work but also for ensuring that their contributions supported the team's overall progress.

On reflection, this project has been a formative experience demonstrating the practical applications of mathematics in an industry setting. It has developed our technical skills, as well as interpersonal skills such as communication and teamwork, that are key to success in industry. Looking forward to our next project, we aim to build on the insights gained from this project, by implementing team progress meetings earlier, using collaborative tools more strategically, and proactively seeking feedback to enhance project outcomes.